

A Room Full of Eyes. A study about social anxiety awareness

by

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Except where explicitly stated all the work in this report, including appendices, is my own and was carried out during my final/fourth year. It has not been submitted for assessment in any other context.

Abstract

The abstract is a bird's eye view of the project.

It should not exceed one page. Mention the scope and objectives of the project, the methodology used, the main findings, and significance of your results.

The page count of 12 is exclusive of bibliography and appendices.

Move above content

Give spoilers about project

Overview Diagram can go after or before abstract Can have multiple

All figures chart, diagram, images must be captions, few words about them e.g. See Figure x

Use insert cross reference.

Acknowledgements

I want to start by expressing my gratitude to Dr Brian Mitchell, who supervised this project. His constant support and guidance were valued and appreciated. This project would not have been achievable without his thorough guidance and fantastic feedback. I would also like to thank my second marker Dr Gennaro Imperatore for all of his insightful comments and feedback on the project presentation, which helped steer and improve the report

Glossary

Lerp - With this function you can find the value that equates to the position between two other values for a given percentage.

Social battery - Refers to 'social exhaustion' and correlates to the HP bar of the game

Headphone battery - Refers to 'music therapy' and correlates to healing in the game

Aspects of social anxiety - Many different types of social anxiety exist, too much to cover in this project. Cognitive and Behavioural aspects are what is covered in the project.

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Have a list of figures, auto generated with figures once made, list of tables instead of figures. ToC, list of figures, list of tables. Glossary just after ToC to explain terms like lerp, social/headphone batteries. Definition of term, how it is used in the project, simple and limitation of full definition. Many aspects of social anxiety can also go in glossary to explain only some of it will be cover

1. Introduction

Although mental health awareness has increased in recent times, disorders like social anxiety disorder are still often incorrectly identified and hard to explain to others who do not suffer from them. Even though standard treatments like therapy give useful support, not everyone can access them. Meanwhile, video games have developed more than just enjoyment, providing interactive settings that can simulate real-world situations in a safe and accessible setting. Very few games explicitly address social anxiety as a core part of gameplay. By creating a game that uses elements of social anxiety through mechanics like social interaction, emotional exhaustion, and coping mechanisms like music-based activities, this project aims to close this gap.

1. Key problems of raising awareness of social anxiety

Social anxiety disorder is an internal condition, making it difficult to raise awareness of it. It can be hard to tell what the effects of social anxiety, which is sometimes confused with shyness. This makes raising awareness much more challenging. Text-based or professional explanations are often used to educate the public about social anxiety, but they could not properly express what it's like to have social anxiety disorder. It can be difficult to accurately show social anxiety while keeping accessibility, which means that original interactive approaches are needed.

2. Importance of the problem

People with social anxiety disorder may experience challenges in their daily lives, including relationships, work, and education. Many cases of social anxiety makes it difficult for people to ask for help because doing so forces them to step outside their comfort zone. Due to this, many who suffer from social anxiety tend to keep it a secret. Social situations that are more encouraging can be set to help raise awareness of social anxiety.

A need for improvement in the methods used to raise awareness of social anxiety. The true nature of social anxiety is not fully explained by common beliefs of the disorder. People find it difficult to relate to these problems as a result, and as a consequence, it goes ignored. To raise awareness of social anxiety, more accessible and approachable communication is required. This might be interactive or experience-based, enabling a larger audience and improving knowledge and impacts of social anxiety.

3. Difficulties of tackling the problem

Because social anxiety is a complicated disorder, using an interactive game to symbolise it might be challenging. Many more aspects to social anxiety exist, including behavioural, emotional and cognitive responses, according to research. Difficulties arise when trying to precisely and completely simulate these in a straightforward game. Raising awareness of social anxiety can only be achieved by including all of these behaviours into basic game mechanics, the social battery system used in this project covers only a small amount of social anxiety.

Additionally, creating a balance between playability and realism is difficult. Although the game developed for this project attempts to mimic some aspects of social anxiety, it must contain entertaining and engaging mechanics without being a misrepresentation of social anxiety. The user experience may suffer if the game is too realistic, and if it is too simplistic, there might be no more representation of the condition. Adding coping mechanisms, like music-based treatment, increases the importance of keeping a balance while making sure that they are accurate in the game and demonstrates the impact of the treatment.

4. Project limitations

The project only addresses some aspects of social anxiety rather than all of its parts. Through the game's social battery mechanic, which runs out during social interaction, this project mainly addresses social exhaustion. The game uses music-based treatment as a coping mechanism, which the player visualises by putting on headphones to shut themselves off from their surroundings. These are simplified methods of observing specific aspects of social anxiety.

This project does not address most of the psychological, behavioural, and environmental aspects of social anxiety, which is a complex condition. The only therapeutic approach available in the game is music-based therapy, far too many coping mechanisms exist for a wide range of conditions that this game does not address. Containing the game's scope to enable it supports the project's goals and objectives results in an interpretive rather than a detailed simulation.

5. Research questions

RQ1: How can a game be designed to represent key aspects of social anxiety in an accessible way?

RQ2: How can a game be designed to represent key aspects of social anxiety in an interactive way?

RQ3: How can gameplay mechanics simulate social situations that reflect the experiences of individuals with social anxiety?

RQ4: How can level design be structured to progressively introduce more challenging social interactions?

RQ5: How can a resource-based mechanic, such as a social battery, represent emotional exhaustion during social interaction?

6. Aims and objectives

Aims

1. To create a game which shows the underlying health issues with social anxiety
2. Create a fully functional game
3. Show ways to cure social anxiety through a progressive story
4. To simulate real social situations to help understand the challenges people face with social anxiety
5. To promote coping mechanisms and positive behavior patterns through gameplay
6. Ensure the game is accessible, interactive and user friendly

Objectives

1. Create a start menu, pause menu and ending
2. Create a character the user can control and use to interact with NPCs
3. Create a level where you can interact with NPCs
4. Implement three levels depicting challenging social scenarios
5. Conduct user testing to gather feedback on gameplay experience
6. Use feedback and analyse it to further improve – possible aim
7. Have functioning 'Social Battery' and 'Headphone Battery' meters

2. Background

This project showcases a 2D game developed using [GameMaker](#). The game consists of three progressive levels, all of which show more challenging social settings. Each level consists of planner social interaction objectives or 'quests' that simulate typical anxiety inducing situations.

Two core mechanics help visualise the experience: a 'social battery' representing cognitive energy depletion during interaction and a headphone battery mechanic linked to music therapy, a newly researched coping strategy. This section reviews the technological, psychological, and design literature that informed these decisions.

Each subsection takes the following pattern. The topic (Game engine, assets) introduces essentials first, what it is, why we need it. Criteria, categorise by essential and desirable. List of candidates, each candidate brief intro, 1-2 sentences explaining it + ref to main website/documentation. Evaluate candidates against criteria. Either by candidate indicates how it makes criteria or not. Or by criterion done as table, nested list, text. Can turn into sentences and make into a small paragraph at the end of background, may not need it.

1. Guidelines for what social anxiety is

Social anxiety disorder (SAD), also referred to as social phobia, is characterised by persistent fear and avoidance of social situations due to concerns about negative evaluation by others [1]. Individuals experiencing SAD often report intense anxiety in situations such as speaking in front of an audience, meeting unfamiliar people, or participating in social gatherings [1]. This anxiety can notably interfere with daily functioning, including educational participation, workplace performance and the development of interpersonal relationships [1].

SAD is considered a common mental health disorder, affecting approximately 5-10% of the global population [2]. Symptoms often emerge before early adulthood and may vary in severity, typically being classified as mild, moderate or severe depending on the level of distress and avoidance behaviour experienced by the individual. While some individuals may experience anxiety only in specific situations, others may experience anxiety across a wide range of social interactions. The causes of SAD remain an active area of research, with current studies suggesting that a combination of genetic, environmental and stress related factors contribute to its development. Additionally, anxiety responses are associated with several brain regions involved in emotional processing, making the disorder complex to study and treat.

Some largely recommended treatments for SAD are cognitive behavioural therapy (CBT). CBT focuses on helping individuals recognise and challenge negative thought patterns while gradually confronting anxiety-provoking situations. A key technique used within CBT is graded exposure, where individuals are progressively introduced to feared social scenarios in

a controlled and supportive environment to reduce avoidance behaviours and build confidence in social interactions [3].

These therapeutic principles informed the design of the present game. The progression of social interactions through quests and NPC dialogue reflects elements of graded exposure, enabling players to engage with increasingly challenging scenarios within a controlled digital environment. By modelling social interaction and emotional fatigue through gameplay systems such as the social battery mechanic, the game aims to simulate aspects of anxiety management strategies in an interactive format.

2. Guidelines for tackling social anxiety

A range of therapeutic approaches have been explored for managing symptoms of social anxiety disorder, including cognitive behavioural strategies, exposure-based interventions and complementary therapies such as music-based treatment. Music therapy has increasingly been recognised as a practical and adaptable method for reducing anxiety symptoms. Recent findings suggest that music therapy can provide psychological benefits that can improve emotional regulation, independence and overall quality of life for individuals experiencing anxiety disorders [4]. Research also indicates that music can influence brain regions associated with emotional processing and mood regulation, contributing to reductions in anxiety, agitation and discomfort. However, the effectiveness of music-based interventions can depend largely on the type of music used and the context in which it is applied, further research is still required to fully understand its long-term physiological effects [4].

Music therapy can be broadly categorised into receptive, active and integrative approaches. Active methods involve participation in music-making activities such as singing or playing instruments, whereas receptive techniques involve listening to selected music as a form of emotional regulation [5]. Studies have suggested that receptive music therapy can be particularly effective in reducing anxiety related to emotional stress rather than performance-based anxiety [6]. Because receptive approaches require relatively low effort from the participant while still offering calming and regulatory effects, they provide a suitable model for representing coping strategies within interactive systems.

Social interaction can be mentally and emotionally demanding for individuals experiencing social anxiety, often resulting in feelings of fatigue or emotional exhaustion following prolonged interaction. To represent this concept within the game, a “social battery” mechanic is used to model the gradual depletion of emotional energy during conversations with NPCs. This system provides a simplified representation of the psychological effort associated with social engagement.

These findings informed the design of the coping mechanics within the game. Players can restore their social battery by activating headphones, representing the use of music as a self-regulation strategy. The headphone mechanic is intentionally limited by its own battery source, introducing a strategy element that reflects the reality that coping strategies provide temporary support rather than a permanent solution. Research into simulated social environments suggests that controlled digital scenarios can help individuals explore anxiety triggers and responses within a safe and structured setting [7]. The game thus combines

music-based regulation with gradually increasing NPC interactions and a social battery mechanic to model fatigue and coping behaviour within an interactive environment.

3. Inspiration

Undertale is an iconic RPG where players can choose to either fight or befriend monsters. These choices influence one of three possible endings (Pacifist, Neutral or Genocide) demonstrating the game's distinctive storytelling and mechanics. Undertale was created by Toby Fox and released on September 15, 2015. The game was developed entirely using GameMaker, which contributed to its relevance as an inspiration for this project, as it demonstrates the capabilities of the same development environment used in this work. Undertale's pixel-based visual style and dialogue-driven gameplay were key influences on the design of the present game.

Stardew Valley is another well-known RPG where players inherit their grandfather's farm and gradually build relationships with members of the surrounding village community while developing their farm. Stardew Valley was created by Eric Barone and released on February 26, 2016. Stardew Valley was created using Microsoft's XNA, a high-level collection of APIs for game development. The game's pixel art style and dialogue-based interactions were the primary aspects that inspired this project. Stardew Valley's pixel art often gets praised for its beauty and ability to express the mood of the game. The game's artwork is popular among many gamers due to its calming features and cosy environment. The game's themes of farming, bonding and community show up in the pixel art, which is both visually pleasing and emotionally impactful. As a result, Stardew Valley serves as an inspiration for pixel art.

4. Game engine selection

A software framework created to make the creation of video games easier is called a game engine. It offers a collection of tools and libraries that manage various elements of game development, including user interaction, artificial intelligence, physics, graphics, and music.

[Unity](#) supports 2D, 3D, virtual reality and augmented reality development. It uses C# as its primary scripting language and is widely used in both independent and commercial development. Unity has a large community enabling for many up-to-date tutorials, easy to find assets and support. However, its wide feature set and structural flexibility are mainly optimised for bigger projects. For smaller 2D projects this could add unwanted complexity.

[Godot](#) is an open-source engine supporting both 2D and 3D development. While flexible and easy to understand, its smaller community compared to Unity and GameMaker can result in fewer learning resources and weaker support. Additionally, as an actively open-source platform, frequent updates may require adjustments to project configurations and scripts during development

[Unreal Engine](#) is widely recognised for high-fidelity 3D rendering and advanced visual capabilities. Blueprint, a visual scripting system intended to streamline development processes, is a part of it. Unreal Engine's structure and feature set focus mainly on graphically intensive and large-scale projects, despite the fact that Unreal Engine provides learning tools

and community assistance. This level of depth would go beyond what is necessary for a simple 2D game.

1. Justification

GameMaker was selected according to the following criteria: suitability for 2D development, ease of prototyping, integrated sprite creation, animation tools, collision systems, room editors and appropriateness for small scale game development. GameMaker uses its own scripting language, GameMaker Language (GML), which has a syntax comparable to JavaScript. GameMaker enables fast prototyping and centralised tools.

The project's goals were closely linked with its focused 2D pipeline, linked asset tools and support for quick prototyping. A simplified programming environment was used because the game's main focus is on simulated social anxiety methods, like the social battery and coping strategies rather than advanced visual techniques. As a result GameMaker minimised needless complexity while offering appropriate functionality.

5. Assets

The assets used in the game were a combination of original creations and externally sourced materials. Original sprites and interface elements were developed using the integrated tools within GameMaker. Additional assets were sourced from itch.io, where all selected materials were distributed under licences permitting free use and modification. Licence terms were reviewed prior to integration.

This mixed approach enabled quick development within the project timeframe. Imported assets were modified where necessary to comply with the overall visual and gameplay requirements. Combining self-created and licensed assets reflect common practice in independent game development.

6. Tutorials used for game engine

GameMaker has a large and active online community that provides extensive support through documentation, code examples and tutorial content. During the development of this project, online tutorials were used to assist with understanding certain programming concepts and engine specific features, in particular, several instructional videos from the YouTube channels of Peyton Burnham [8] and the official GameMaker channel [9] were referenced to support implementation of gameplay systems and development workflows.

3. Specification and design

a. Method

This project used [Agile](#), a software development process which emphasises iterative development, adaptability and continuous improvement through the project lifecycle. In an

Agile environment, you are divvying up work into Sprints, which are time-based bursts of activity, typically one-to-four weeks in length. This cadence dictates how much work is completed during a given period[10]. This method enabled the project to accommodate evolving design requirements without notably disrupting the set timeline. Given the relatively limited scope of the game compared with large scale commercial productions, Agile was deemed more suitable than linear methods such as the Waterfall model. Waterfall is particularly useful for large, complex projects with very specific and unchanging requirements. Development teams will be less resistant to detailed product requirements documents and design specifications since that's what's expected[10]. The flexibility inherent in Agile supported incremental development, regular evaluation and ongoing refinement of both gameplay mechanics and technical implementation.

2. Analysis on turning social anxiety into a game

The research on social anxiety and coping strategies covered in the previous chapter served as the inspiration for the software product. People with social anxiety disorder frequently indicate that social contacts are a major cause of stress, which frequently leads to avoidance behaviour and emotional exhaustion.

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People with social anxiety disorder frequently indicate that social contacts are a major cause of stress, which frequently leads to avoidance behaviour and emotional exhaustion. The project's goal was to create a game that could mimic social interactions in a regulated and simplified virtual world.

Both technological and design factors had a role in the decision to create a 2D narrative-driven game. While maintaining interactive dialogue systems and character interactions, a 2D format enables more straightforward visual representation and reduced development complexity. This strategy also fits with the features of selected development environment, GameMaker, which is especially well-suited for creating 2D games.

By analysing the psychological ideas covered in the literature study, important gameplay mechanisms were found. The social battery mechanic, which symbolises the player's available social vitality during gameplay, was implemented based on the idea of emotional exhaustion after social contact. The headphone feature, which enables players to replenish their social energy, was also inspired by studies on coping mechanisms including music-based relaxation.

In an interactive setting, these mechanics were intended to depict simpler parts of anxiety control.

3. Requirements

a. Functional Requirements

- 1 The basic requirements for core gameplay functionality are:
 - 1.1 Player must be able to move in 4 directions (up, down, left, right) using arrow keys.
 - 1.2 Player sprite must animate based on direction of movement
 - 1.3 Player must stop when colliding with walls or obstacles
 - 1.4 NPCs must detect when the player is within interaction range
- 2 The requirements for the social battery system are:
 - 2.1 Player must have a social battery that starts at 100 and displays in the UI
 - 2.2 Social battery must decrease by a fixed amount when talking to an NPC
 - 2.3 Social battery must be clamped between 0 and 100
 - 2.4 When social battery reaches 0 the player must lose
- 3 The requirements for the headphone system are:
 - 3.1 Player must have a headphone battery that starts at 100 and displays in the UI
 - 3.2 Player must hold 'E' to activate headphones
 - 3.3 When headphone charge reaches 0 headphones cannot be used
 - 3.4 Player must be unable to move during headphone animation
- 4 The requirements for the dialogue system are:
 - 4.1 Dialogue must support multiple messages in sequence
 - 4.2 Dialogue box must display character name for each message
- 5 The requirements for the level and progression system are:
 - 5.1 System must support level progression
 - 5.2 Each level must maintain social and headphone battery states
 - 5.3 System must track objective completion
 - 5.4 Completing all objectives across all levels must trigger win condition
 - 5.5 Player must complete all level objectives to progress

2. Non-Functional Requirements

1 The requirements for performance are:

- 1.1 Social battery lerp must be smooth and responsive
- 1.2 Camera/viewport scaling must not impact performance

2 The requirements for usability are:

- 2.1 UI elements must be clearly visible and readable
- 2.2 Social and headphone battery must always be visible
- 2.3 Controls must be simple and consistent throughout the game
- 2.4 Interaction cues must be visible when prompted
- 2.5 Dialogue text must be easy to read with proper spacing and formatting

3 The requirements for maintainability are:

- 3.1 Code must use parent-child object inheritance to reduce code duplication
- 3.2 Global variables must be properly initialised in a centralised location

4 The requirements for scalability are:

- 4.1 Dialogue system must support unlimited messages per conversation
- 4.2 Branching dialogue must support unlimited choice options
- 4.3 System must support adding new NPCs without modifying core code
- 4.4 Game must support multiple rooms with varying dimensions
- 4.5 Battery cost and values must be easily adjustable

5 The requirements for visual design are:

- 5.1 Player sprite must have distinct animations for all four directions
- 5.2 Dialogue box must occupy appropriate screen space
- 5.3 UI elements must have consistent pixel art styles with game aesthetic
- 5.4 Depth sorting must ensure proper visual layering of all objects

6 The requirements for educational/experiential value are:

6.1 Objective system must simulate social situations that cause anxiety

6.2 Game mechanics must represent experiences with social anxiety

6.3 Headphones must represent a realistic coping mechanism

4. Design

a. Interface Design

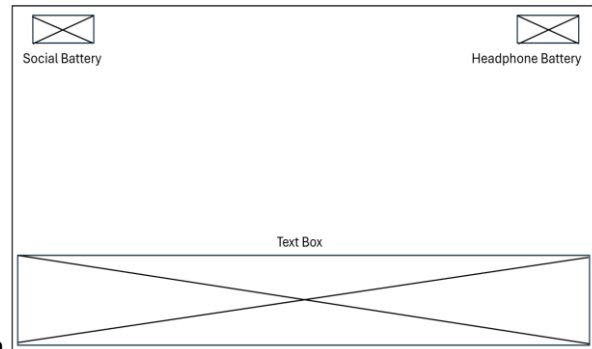


Figure 1. Wireframe layout of the game showing social battery, headphone battery and text box.

The user interface for the game was designed to ensure the important gameplay information remains visible and easy to understand. To assist with the process, Figure 1 was created to explore the layout and placement of key interface parts before implementation. The wireframe helped visualise how elements such as the social battery, headphone battery and dialogue box would appear on screen during gameplay.

Typically Health Point bars are placed in corners of the screen so that the user's view is not obstructed from the game. I decided to position these in the top corners of the screen as it enables the player to quickly monitor their resources and doesn't interfere with gameplay. These systems represent important gameplay mechanics, as the headphone battery controls the use of the music-based coping mechanism while the social battery reflects the player's current emotional energy.

The dialogue box was positioned along the bottom of the screen to enable conversations with non-playable characters to be neatly displayed without blocking environment visibility. Similar to Undertale and Stardew valley this is common interface conventions used in many 2D role-playing games, ensuring that dialogue remains readable while maintaining focus on the game world.

Figure 1 was evaluated through iterative testing during development. After implementing the interface within the game engine, adjustments were made to space, scaling and placement to ensure that user interface elements remained visible across different room sizes and gameplay situations. This process helped refine the final interface design while maintaining clarity and usability for the player.

3.4.2 System Design

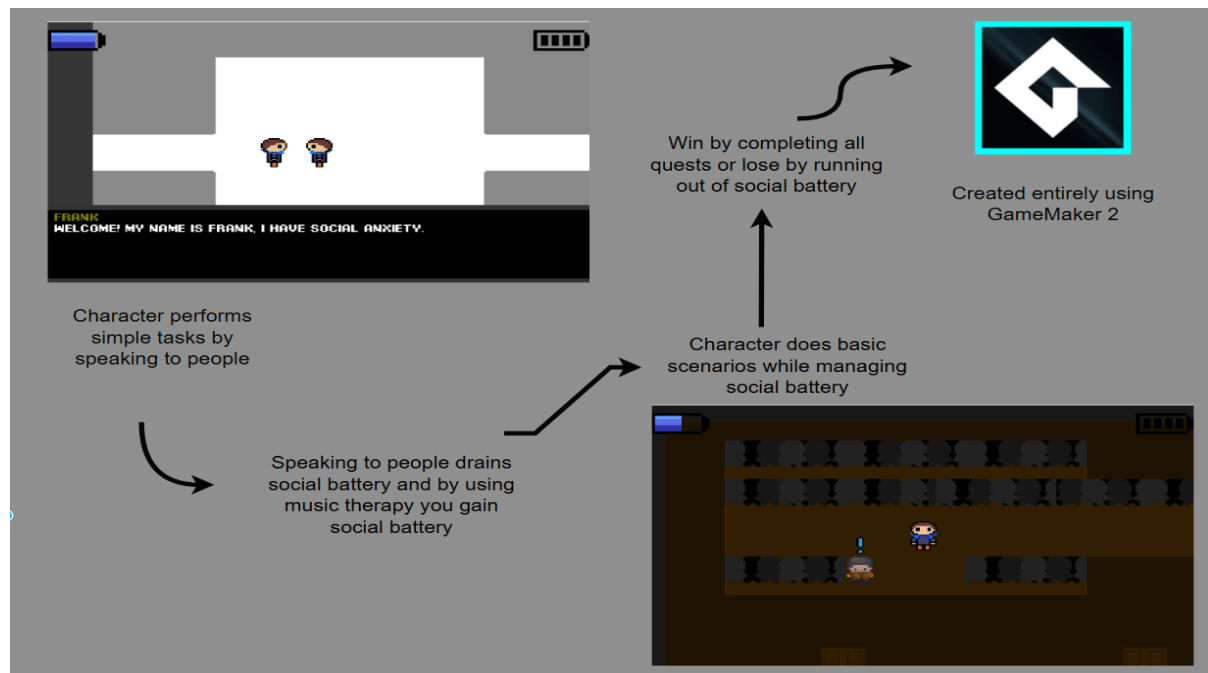


Figure 2. Overview diagram created to help benefit game creation.

To show the structure of the game's main systems and how they work together while playing, Figure 2 was created. The player advances through the game by finishing quests that require interaction with NPCs. The main resource management mechanism in the game is the player's social battery which is reduced by a set amount by each interaction with another NPC. The player may deal with this by using headphones to restore energy, which represents music as a coping strategy. Each level's goal is to finish all interactions while preserving social battery. The player wins the level if they successfully complete every objective and the player loses if the social battery runs out.

4. Product

a. Implementation

GameMaker, a game engine mainly used for 2D game development, was used to create the game. Fast prototyping and testing throughout production are made possible by GameMaker's integrated development environment, which includes room editors, scripting features and sprite editing and animation tools. GameMaker's own programming language, GameMaker Language (GML), which resembles Javascript, is used. The game's core parts, such as player movement, non-player characters (NPCs) interaction and the social and headphone battery systems used throughout, were created using GML.

Using GameMaker's object system, the system architecture is simple and object-orientated. The player object serves as the focal point of the game and communicates with other elements including dialogue controllers, NPCs and battery management systems. To minimise code duplication and enhance maintainability, parent-child object inheritance was employed. For instance, particular dialogue interactions were defined within child objects, whereas common behaviour shared by NPCs was implemented in a parent object.

Since the project is an offline single-player game, no network communication or external servers were needed. No personal user data is gathered, saved, or exchanged, as all gaming mechanisms function locally within the game engine. Therefore, no particular methods for protecting personal data were needed. To ensure stability and stop unexpected system behaviour, development techniques such as an organised code structure and restricted access to global variables were put in place.

2. Verification & validation

Verification and validation steps were used during the development stage to ensure that the game functioned correctly and complied with the requirements listed in section 3.3. Verification concentrated on ensuring that the implemented systems operated as intended during development, whereas validation ensured that the final product met the functional and non-functional requirements stated during the design stage.

After installation, each gameplay system, including player movement, NPC interaction, dialogue advancement and battery management, was examined separately. Variable values were tracked, event triggers were examined and runtime faults were found using GameMaker's debugging tools. For instance, the social battery system was tested to make sure that the battery values stayed within the specified range of zero to one hundred and correctly dropped when interacting with NPCs. Similarly, collision detection was checked to make sure the player couldn't get past objects or walls in the game world.

The finished game was validated by comparing it to the functional and non-functional requirements outlined in section 3.3. To ensure that every feature performed as planned, functional requirements such as player movement, NPC interaction, dialogue progression and level objectives were examined during gameplay. By making sure that UI parts like the social and headphone battery meter, and dialogue boxes remained visible and noticeable throughout gameplay, non-functional needs like usability and interface clarity were assessed. By verifying that the game maintained fluid gameplay without lag or frame drops during testing, performance factors were also validated.

5. User study

a. Evaluation Process

If you involved humans in the evaluation, how many did you have? What can you say about the demographics of your participants? (If you did collect these)

In terms of the user interface, how did you carry out a usability evaluation, how did you go about doing this? How did you recruit participants?

In terms of delivering functionality, did you carry out user acceptance testing? (see attached guidance).

Three user testers were recruited for the project. Emails were sent out to each year group, small advertisement was done in the psychology department and verbal communication around friends and family.

The recruitment from that was sent to participants provided them with guidance and an idea of what to expect during the evaluation process. Unless it was necessary to move forward with the assessment, participants were instructed to play the game independently and experience it for the first time. The game's social and headphone batteries, as well as NPC interactions, were used by testers. This gave testers with good game knowledge to provide constructive feedback.

To ensure that the project followed the initial criteria, user acceptance testing was conducted. To finish three stages of the game, testers had to manage their social and headphone batteries while completing taskings, while interacting with NPCs. This provided positive feedback and validated the claim that the mechanics met the project's requirements.

2. Results of Evaluation

This Section includes a direct interpretation of the gathered data and evaluation processes.

Each participant was given the same question sheet and asked five questions for each level, then finally six questions at the end of the game. Two alternative questions were made for in between levels if the tester got stuck during a level however these were not used. The questions were asked in context of the level the tester had just played up to and including level three.

Level one was realistic according to all the testers, nothing happened that they expected to happen differently and went smoothly. The tutorial just before the first level, which was not

counted as a level, gave clear instructions. All testers said after the first interaction and with the tutorial's guidance the game was easy enough to follow mechanically from that point. For the social battery mechanic one tester said that they were "nervous it could run out since it was a new experience". This was good feedback as it demonstrated how realistic the social battery was for testers, wary and unsure of how to go about it.

Level two was similar in responses to level one. The smaller level made it clear and easy to follow, straightforward instructions and nothing expected to happen differently. The main mechanic of level two was the main character's social battery automatically decreased while in the party, managing this alongside doing quests gave more of a challenge than some testers were expecting. The testers really enjoyed this feature as it made the game feel more realistic, they wanted to leave as soon as possible as they didn't want to run out of social battery. The headphones were effective but weren't enough to fully block the social battery loss out.

Level three was the final level, it was a presentation. Similar results from the last two levels, it was realistic, harder to manage social battery than the last level and easy to follow as instructions were clear. A tester related to this experience quite well as they themselves had been through presentations before and were always nervous when doing them. "I was worried about what to say, I didn't want to skip parts but also had to manage the social battery". This helped shape that the game was realistic, the player's social battery was limiting and stopping them from saying everything which is a common experience for people with social anxiety.

After the game was complete, each tester was asked six questions about the overall experience of the game. The overall experience of the game was similar to 2D games, controls were consistent and no controls changed over time. Every tester said the game helped represent an experience of social anxiety, "it's all a battle of trying to help others while constantly worrying about yourself", was one given response, this shows that the game gets the main point across to the user. It was a self explanatory game, after the first interaction it became simple to understand, making the game accessible and easy means for a large amount of players being able to experience the game. All testers said it would be a good game for both people who experience social anxiety and who want to learn about social anxiety.

3. Returning to the Research Questions

5.3.1 RQ3

The findings from the user study showed that the quests and levels in the project reflect experiences that individuals with social anxiety may encounter in everyday situations. This suggests that the gameplay structure successfully represents social scenarios and therefore fulfils this research question.

5.3.2 RQ1

The findings from the user study showed that the game mechanics and interface helped communicate key aspects of social anxiety in an accessible way. This indicates that the design successfully represents elements of the condition and therefore fulfils this research question.

5.3.3 Objective 2

During the user testing, participants were required to use the in-game character to move and interact with NPCs in order to progress through the game. This demonstrated that the character could be successfully controlled by the player and could interact with NPCs as intended, thus fulfilling this objective.

5.3.4 Objective 5

The user testing included questions such as “what aspects of the game worked well” and “what aspects of the game would you change”. All testers agreed that the game effectively delivered an experience representing social anxiety. One tester noted that in the final level the podium required to begin the presentation was difficult to see, and suggested adding an exclamation mark above it to make it more noticeable.

6. Discussion & Reflection

a. Interpreting the Results

The game’s mechanics were accurate and well-described, according to testers. The testers were able to experience what social exhaustion is like thanks to the social battery feature[2]. The social battery was able to produce pressure, as evidenced by one tester’s observation that the social battery mechanic made them anxious in case it ran out. The project’s goal is supported by the user evaluation findings, which also shows how well the game’s mechanics capture elements of social anxiety.

Every level had a scenario, such as a presentation or a cafe, that the testers found to be realistic. The testers like how the game’s levels got harder as it continued, creating a greater sense of pressure that is a reflection of those who suffer from social anxiety. The headphone’s, which stand in for music therapy[5], also showed how coping strategies could be implemented into gameplay while also being easy to understand.

The findings show how well the interface design, level structure, and game mechanics complemented one another to simulate the intended experience. After the tutorial, testers said the game was easy to comprehend, with very straightforward controls and simple to understand quests throughout the game. This shows how the project successfully balanced accessibility with the representation of a difficult subject, supporting the goal of developing an interactive experience that increases social anxiety awareness.

2. Development Difficulties

This section will explain some of the difficulties encountered during the project. Each subsection will go over the problem, how it happened and finally how it was fixed. These will be made in the context of milestones, achievements that helped progress the project further.

6.3.1 User Interface (UI)

The user interface was simple to design, and as explained in the previous sections the placement in the top corners was so that the player could see the environment. Towards the end of the project during the creation of a pause menu, it was noticeably larger than all other elements in the UI. This happened due to the display size being set in the players code at the beginning for drawing the initial UI and then not being set back which made everything else bigger. To fix this hurdle, the correct display size was set up and stored before putting the display size back into its smaller form to draw the UI. It was then reused again from its stored state so that everything would be back to its similar size.

6.3.2 Dialogue System

Since dialogue is the main focus of this game, the dialogue system needed to be both efficient and user friendly. When NPC dialogue and quests were first created, they were kept inside the respective NPCs code. As more quests and dialogue were implemented, it became much more difficult to keep track of everything. With all the extra code required to keep earlier NPC quests from activating when they shouldn't, the second level became a mess. For each of the three levels, three initialisation objects were made to hold all relevant data. Everything functioned more smoothly when the correct object was active for its level. Simple function calling was created to handle quests in Scripts, these were global across all objects in the project and were developed to contain custom functions like complete or finish a quest.

6.3.3 Save States across levels

Each level in the game was built as a distinct room, requiring objects to keep track of specific game conditions during room transitions. Quests as well as the social and headphone battery values, reset each time the player entered another room during early testing. Using global variables and writing a script to monitor quest progress between rooms were the first attempts to fix the issue. Additional debugging showed every room had the player object, which held the game state logic. When another level was entered it would take the current player object, overwrite it, and reset the states. Making sure the player object only existed in the starting room enabled the states to not reset. The quest progress function was then later removed as it added more needless functions.

3. Limitations

The scope for the project was made larger than the time given for development. There were times throughout development when time became more of an issue than implementation. All fundamental systems and gameplay mechanics described in the original project plan were successful however extensive work and some corner cutting was required for everything to be completed in a timely manner.

To generate ideas for the project, existing games were analysed in section 2.5. The games described there were focusing on gameplay mechanics and environment design, not on social anxiety. Games where social anxiety is the primary aspect aren't as common in current times, a few examples exist but they are either niche or don't have enough acknowledgement to provide good reference material. This both encouraged a greater creative freedom but also hindered the project slightly as ideas related to games using social anxiety had to be all thought up independently rather than being directed by set design approaches.

A lack of project planning made the project development a bumpier road than it should have originally been. The project started with a broad idea for the game, however as implementation began, the gameplay features and level structures were not fully planned out. Rather than being organised and having a straight path, the development became more reactive, with original mechanics and concepts being added as they came to mind. Had everything been thought out originally during the planning phase, development efficiency would have increased drastically and changes to the project wouldn't have needed to happen so often.

The project was developed by a single individual meant fewer opportunities for testing, feedback and problem solving. Larger development teams can help sharing tasks and peer evaluations, which can assist find early problems and eventually boost project quality. Despite the limitations of a single person, the project was completed on time and the intended basic mechanics were completely designed.

7. Conclusions

The conclusion is similar to when a plane lands. You don't rewrite the introduction. You say something like - I addressed the problem outlined in the introduction, and I built some software to do this. I tested the software like this ADD FEW WORDS.

Summarize main findings drawn from the project work. Mention the objectives or research questions. Do not repeat points raised in the discussion and reflection Section. If applicable, you can make recommendations. The conclusion should NOT contain any references.

1. Summary

The goal of this project was to create a game that uses interactive gameplay mechanics to show different aspects of social anxiety. Using GameMaker, a 2D top-down game with necessary parts like player movement, NPC interactions, dialogue system and a quest based progression framework was successfully developed. The headphone mechanic portrays music as a coping strategy for lowering anxiety, the main gameplay feature is the social battery system to which was used to resemble emotional fatigue during social interactions.

Research into social anxiety and treatments, as well as a study of previously explained games that inspired the project's visual art and gameplay structure. These influences helped shape the mechanics, environment design and user interface to make sure the gameplay experience matched the project's goal of simulating social interactions and coping behaviours in a way that was accessible.

During the development process, testing and verification took place to make sure the game systems behaved accordingly and that functional and non-functional requirements worked

as expected. The project's final implementation demonstrates that games can be used to represent psychological scenarios in an interactive way.

2. Future Work

There are multiple ways in which the project could be expanded on. Voice acting and sound effects would be one way to improve the game's emotional depth. Character voices, ambience and background music are a few examples of audio features that could enhance the game.

Moving from a 2D game environment to a 3D game world is a good area of development. 3D would help characters be more expressive and more immersive graphics would help set the mood of the game. The game would greatly benefit being moved into a 3D world setting, emotional elements like social interactions would be greatly improved.

The gameplay can be expanded by adding more quests and harder level designs. Currently only up to three quests exist in some levels, if more levels were to be added the number of quests will also be increased. Levels could also be made more challenging, with harder puzzles and quests that require extra thinking or multiple NPCs to talk to before finding an answer which can push players to explore more of the level. These parts would make the game more engaging and help the game have more depth.

3. Reflections

With all the new information present about GameMaker and game development, the project could have been enhanced if it had been started over. The usage of GameMaker's integrated features is one of the main areas that needed improvement. Some system architectures were created manually during early phases rather than utilising the engines prebuilt features. For example, GameMaker's UI layer features for room customisation. This was more commonly used towards the end of the project however if this was used in the beginning would have saved more time with setting up UI elements. Had these features been used early on, development time would have been shortened.

The development process offered important knowledge and skills in project planning and technical implementation. Planning object structures and utilising current engine features to increase efficiency and robustness would be important if the project were to be developed further.

4. What I would do the same

Despite the lack of fully using GameMaker's features, the features that were utilised during the development process were extremely helpful in the creation of the project. Creating sprites internally made it easy to design and modify visual elements without needing external software. This enabled changes to be made quickly during development, which was useful when adjusting character animations and interface elements to better match the overall design of the game.

Tutorials were used to learn about new features in the game engine in order to get a head start on the project. When structuring fundamental gameplay features, the tutorials served as a solid start for understanding how various mechanics may be implemented. These tutorials sped up the development of the social battery mechanism, dialogue systems and NPC interactions

5. Final Thoughts

The project effectively achieved its main goals and serves as a base for future work, showing how games can explore mental health-related topics while keeping engaging gameplay elements.

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A. Appendix

This is where you can include your documentation.

Remember that the marker is not required to read this, but might well check to ensure that you have included product documentation, and ethical approval, as required.

1. Ethical Approval Form

If your project required you to do any evaluation with humans, you **MUST** include this. It can be downloaded from the Ethics system.

<https://local.cis.strath.ac.uk/wp/extras/ethics/index.php?view=3078>

The link above is my ethics approval form.

2. Participant Information Sheet

If your project required you to do any evaluation with humans, you **MUST** include this

<https://www.strath.ac.uk/ethics/information-sheet-and-consent-form/>

3. Consent Form

If your project required you to do any evaluation with humans, you **MUST** include this.

<https://www.strath.ac.uk/ethics/information-sheet-and-consent-form/>

B. Engineering Requirements

This is all the functional and non-functional requirements

Number instead of bullet

1. Functional Requirements

Player character Movement and Controls:

- Player character must be able to move in 4 directions (up, down, left, right) using arrow keys
- Player character Sprite must animate based on direction of movement
- Player character must stop when colliding with walls or obstacles
- Player character depth must be adjusted based on Y position for proper layering

Social Battery System:

- Player character must have a social battery that starts at 100 and displays in the UI

- Player character must have a headphone battery that starts at 100 and displays in the UI
- Social battery must decrease by a fixed amount when talking to an NPC
- Social battery must be clamped between 0 and 100
- When social battery reaches 0 the player must lose

Headphone System:

- Player must hold 'E' to activate headphones
- When headphone charge reaches 0 headphones cannot be used
- Player must be unable to move during headphone animation

NPC Interaction System:

- NPCs must detect when the player is within interaction range
- Parent NPC object must allow inheritance for shared behavior
- Each NPC must have configurable social battery cost

Dialogue System:

- Dialogue must support multiple messages in sequence
- Dialogue box must display character name for each message
- System must preserve GUI scaling for other UI elements during dialogue

Game Initialisation:

- Game must initialise all global dialogue variables before NPCs are created
- GUI size must be set appropriately for camera/viewport configuration
- Character color maps must be defined globally

Level System:

- Game must support three different levels
- System must support level progression
- Each level must maintain social and headphone battery states
- System must track objective completion
- Completing all objectives across all levels must trigger win condition
- Player must complete all level objectives to progress

2. Non-Functional Requirements

Performance:

- Social battery lerp must be smooth and responsive
- Camera/viewport scaling must not impact performance

Usability:

- UI elements must be clearly visible and readable
- Headphone battery must be always visible

- Social battery must be always visible
- Controls must be simple and consistent throughout the game
- Interaction cues must be visible when prompted
- Dialogue text must be easy to read with proper formatting
- Dialogue text must not overlap with one another

Maintainability:

- Code must use parent-child object inheritance to reduce code duplication
- System must use macros for commonly used values
- Global variables must be properly initialised in a centralised location
- Using accurately descriptive names for identifiers

Scalability:

- Dialogue system must support unlimited messages per conversation
- Branching dialogue must support unlimited choice options
- System must support adding new NPCs without modifying core code
- Game must support multiple rooms with varying dimensions
- Battery cost and values must be easily adjustable

Accessibility:

- Font must have anti-aliasing off for pixelated clarity
- Font must have legibility
- UI must adapt to different camera/viewport configurations
- Text must have sufficient contrast against dialogue box background

Visual Design:

- Player sprite must have distinct animations for all four directions
- Dialogue box must occupy appropriate screen space
- UI elements must have consistent pixel art styles with game aesthetic
- Depth sorting must ensure proper visual layering of all objects

Educational Value:

- Objective system must simulate social situations that cause anxiety
- Game mechanics must represent experiences with social anxiety
- Headphones must represent a realistic coping mechanism
- Social battery depletion must create tension in comparison to real anxiety

Code Quality:

- Variable names must be descriptive and follow camelCase convention
- Internal commentary must explain complex logic and system behavior
- Event code must be organised logically
- Magic numbers must be avoided

